

Lab 7 - Inference on Means

North Carolina Births

In 2004, the state of North Carolina released to the public a large data set containing information on births recorded in this state. This data set has been of interest to medical researchers who are studying the relation between habits and practices of expectant mothers and the birth of their children. We will work with a sample of observations from this data set. These 999 cases were chosen at random.

Let's load the `nc` data set into our workspace. The data is in a file called `nc.csv`.

```
# change working directory to the folder
# where the data is saved
nc = read.csv("nc.csv", header=T)
# attach dataset to workspace
attach(nc)
```

We have observations on 13 different variables, some categorical and some numerical. The meaning of each variable is as follows.

<code>fage</code>	father's age in years.
<code>mage</code>	mother's age in years.
<code>mature</code>	maturity status of mother.
<code>weeks</code>	length of pregnancy in weeks.
<code>premie</code>	whether the birth was classified as premature (premie) or full-term.
<code>visits</code>	number of hospital visits during pregnancy.
<code>gained</code>	weight gained by mother during pregnancy in pounds.
<code>weight</code>	weight of the baby at birth in pounds.
<code>lowbirthweight</code>	whether baby was classified as low birthweight ('low') or not ('not low').
<code>gender</code>	gender of the baby, 'female' or 'male'.
<code>habit</code>	status of the mother as a 'nonsmoker' or a 'smoker'.
<code>marital</code>	whether mother is 'married' or 'not married' at birth.
<code>whitemom</code>	whether mom is 'white' or 'not white'.

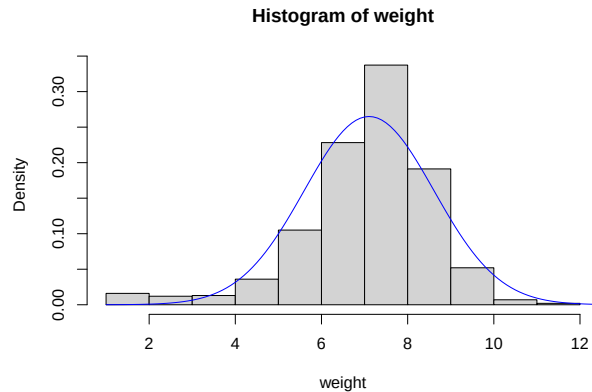
First, let's calculate the mean and the standard deviation of babies' weights in this sample as well as the sample size.

```
xbar <- mean(weight)
s <- sd(weight)
n <- length(weight)
```

Exercise 1: Make a histogram of weight of the baby, `weight` variable, and describe the distribution of this variable.

Now we make a density histogram to use as the backdrop and use the `lines()` function to overlay a normal probability curve. The difference between a frequency histogram and a density histogram is that while in a frequency histogram the **heights** of the bars add up to the total number of observations, in a density histogram the **areas** of the bars add up to 1. The area of each bar can be calculated as simply the height $\times$ the width of the bar. Using a density histogram allows us to properly overlay a normal distribution curve over the histogram since the curve is a normal probability density function. Frequency and density histograms both display the same exact shape; they only differ in their y-axis. You can verify this by comparing the frequency histogram you constructed earlier and the density histogram created by the commands below.

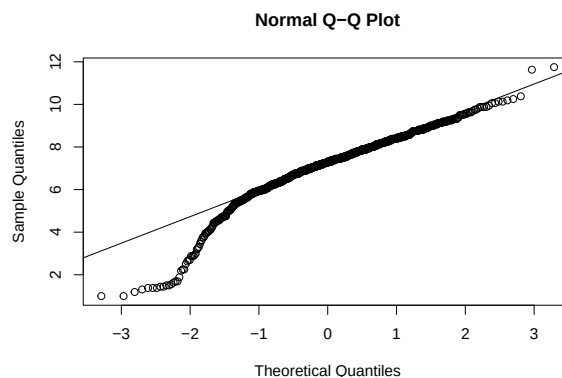
```
# Plot histogram
hist(weight, probability = TRUE)
# Weights range
x = seq(1, 13, by=0.1)
# Compute density for x values
y = dnorm(x = x, mean = xbar, sd = s)
# Draw line on top of histogram
lines(x = x, y = y, col = "blue")
```



After plotting the density histogram with the first command, we create the x- and y-coordinates for the normal curve. We chose the `x` range as 1 to 13 in order to span the entire range of `weight`. To create `y`, we use `dnorm()` to calculate the density of each of those `x`-values in a distribution that is normal with mean `xbar` and standard deviation `s`. The final command draws a curve on the existing plot (the density histogram) by connecting each of the points specified by `x` and `y`.

Eyeballing the shape of the histogram is one way to determine if the data appear to be nearly normally distributed. A normal probability plot or Q-Q plot is also a good alternative approach to check for normality.

```
qqnorm(weight)
qqline(weight)
```



Exercise 2: Based on these plots, does it appear that the data follow a nearly normal distribution?

Note that even though the Q-Q plot displays a deviation from normality, the sample size is large enough ($n = 999 > 30$) such that the results from the Central Limit Theorem still apply and we can proceed to perform inference on the parameters of interest.

Confidence Intervals for a Mean

If we want to estimate the average weight of a baby at birth, we can use information from this sample to build a confidence interval for the true population parameter.

Remember that a confidence interval is always of the form

$$\text{point estimate} + ME$$

In this case the point estimate is the sample mean. The margin of error is always composed of two components, a critical value (t^* or z^*) and a standard error. The standard error for the sample mean can be calculated as $\frac{s}{\sqrt{n}}$. Let's calculate that first.

```
se <- s / sqrt(n)
```

Note that for $n = 999$, the t-distribution – which is the exact distribution to use to calculate a confidence interval for the mean – is indistinguishable from the normal. By now, you probably know that z^* for a 95% confidence interval is 1.96. This value comes from the fact that $z = -1.96$ and $z = 1.96$ are the cutoff values for the middle 95% of a standard normal distribution. In other words, they are the cutoff points for the 2.5th percentile (in the lower end) and $2.5 + 95 = 97.5^{\text{th}}$ percentile (in the upper end). In order to get R to calculate this value for us, we use

```
qnorm(0.975)
```

```
## [1] 1.959964
```

So if we wanted to calculate the critical value for a 98% confidence interval, we first need to realize that the middle 98% of the distribution will be bounded by the 1st and the 99th percentiles, and hence the critical value will be

```
qnorm(0.99)
```

```
## [1] 2.326348
```

since the critical value is always positive.

Exercise 3: Verify that the critical values above are very similar to those obtained from the corresponding t-distribution.

Let's get back to the calculation of the confidence interval. If we want to construct a 95% confidence interval we can set z^* to be

```
zstar <- qnorm(0.975)
```

The margin of error is simply the product of z^* and the standard error

```
me <- zstar * se
```

Then, the bounds of the confidence interval can be calculated as

```
xbar - me
```

```
## [1] 7.011111
```

```
xbar + me
```

```
## [1] 7.197838
```

Exercise 4: Interpret the confidence interval. Make sure to put it in context of the data.

Hypothesis Testing for Means

In first world nations, the average birth weight of a full-term new born is approximately 7.5 pounds. Let's see how the average weight of babies born in North Carolina compare to this value.

In order to do this we need to first subset our dataset to include only full-term babies. We'll call this new data set `nc_ft`.

```
nc_ft <- subset(nc, nc$premie == "full term")
```

You should see a new data set appear in your workspace which has 846 rows and 13 variables. Earlier we saw that there were 152 premature and 846 full term births, hence we know that there should be 846 cases (rows) in the new data set that we just created.

Let's calculate the mean birth weight for the full term babies

```
xbar = mean(nc_ft$weight)
boot_means <- replicate(5000, mean(sample(nc_ft$weight, replace = TRUE)))
```

This value is slightly below 7.5 pounds, but it's close.

Exercise 5: How would you expect the sample mean of birth weight to differ between the complete dataset (nc) and the subsetting dataset (nc_ft)?

Exercise 6: Write the hypotheses for testing if the average weight of full term babies born in North Carolina is different than 7.5 ounces.

To conduct the hypothesis test we need to first calculate a test statistic, a z-score, using the formula

$$z = \frac{\bar{x} - \mu}{SE}$$

```
mu <- 7.5
s <- sd(nc_ft$weight)
n <- length(nc_ft$weight)
se <- s / sqrt(n)
z <- (xbar - mu) / se
```

Calculating the p-value is a two step process: first find the tail area, and then multiply by two since the alternative hypothesis is two sided.

```
tail <- pnorm(z, lower.tail = TRUE)
pval <- tail * 2
```

Note that we used `lower.tail = TRUE` since the z-score we obtained was negative. If we wanted to get the upper tail, which is often the case when we have a positive z-score, we would use `lower.tail = FALSE`.

Exercise 8: What is the conclusion of the hypothesis test? Make sure to provide an interpretation in context.

More Practice

1. Using the original data set, `nc`, create a new data set called `nc_sm` that only contains information from smoker mothers. Then calculate the average weight gained by these mothers.

Note that the variable 'gained' contains missing values that will produce a value of `NA` (i.e. Not Available) in your calculations. To remove the missing values (after you create the `nc_sm` data set), define a new variable called `gain` using the following code :

```
gain <- nc_sm$gained[is.na(nc_sm$gained)==F]
```

Use the `gain` variable you just created to perform all calculations.

2. Check if the assumptions and conditions necessary for inference on a mean are satisfied for this sample of smoker mothers.
3. Construct a 95% confidence interval for the average weight gained by smoker mothers and provide an interpretation for this interval in context.

4. Weight gain of roughly 30 pounds is considered normal. Conduct a hypothesis test to evaluate if the average weight gain of smoker mothers is different than 30 pounds. Make sure to interpret your conclusion in context of the question.
5. Do the results of the confidence interval and hypothesis test agree with each other? Is this what you would expect or not? Explain.

Inference for proportions

Follow similar steps for the following exercise on inference for proportions i.e. check assumptions, show your calculations, state the hypothesis, and interpret your results in context.

6. Estimate the population proportion of premature births p using a 95% confidence interval.
7. Calculate a 95% bootstrap confidence interval for the proportion of premature births. How does it compare to the interval in question 6?
8. In the United States, the proportion of live births that are premature is approximately 10%. Test the hypothesis that the proportion of premature births is greater than 10%.